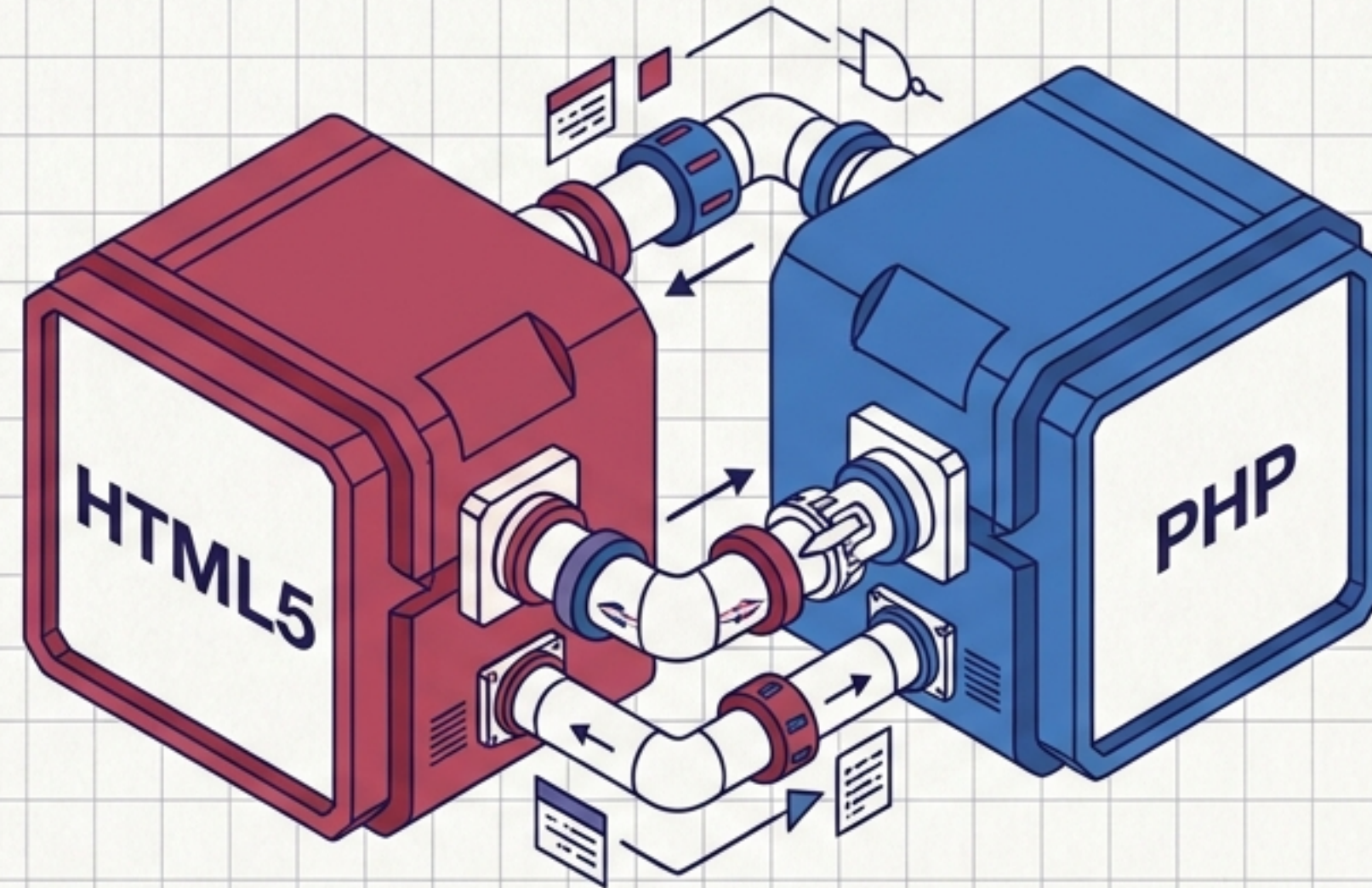
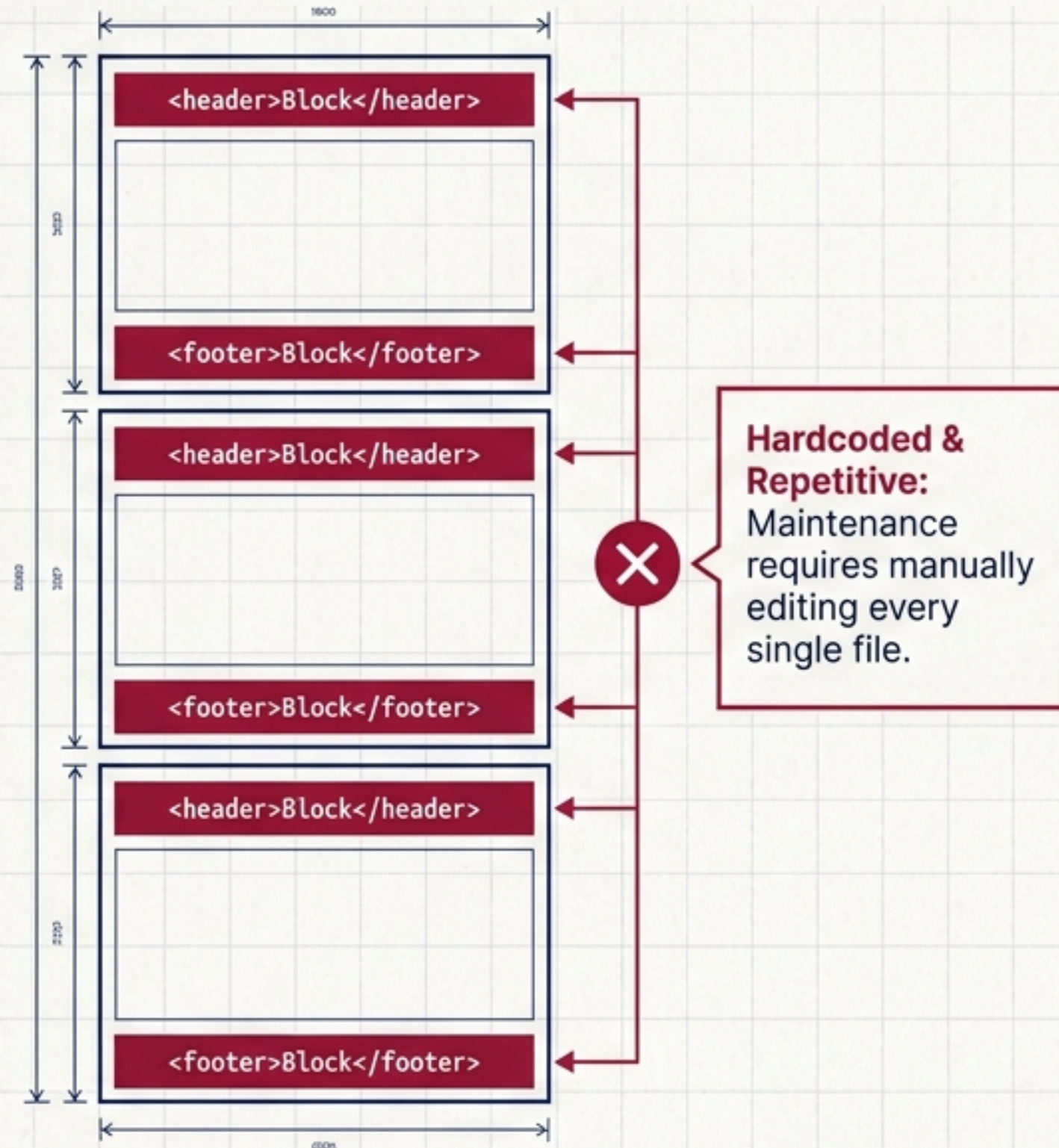




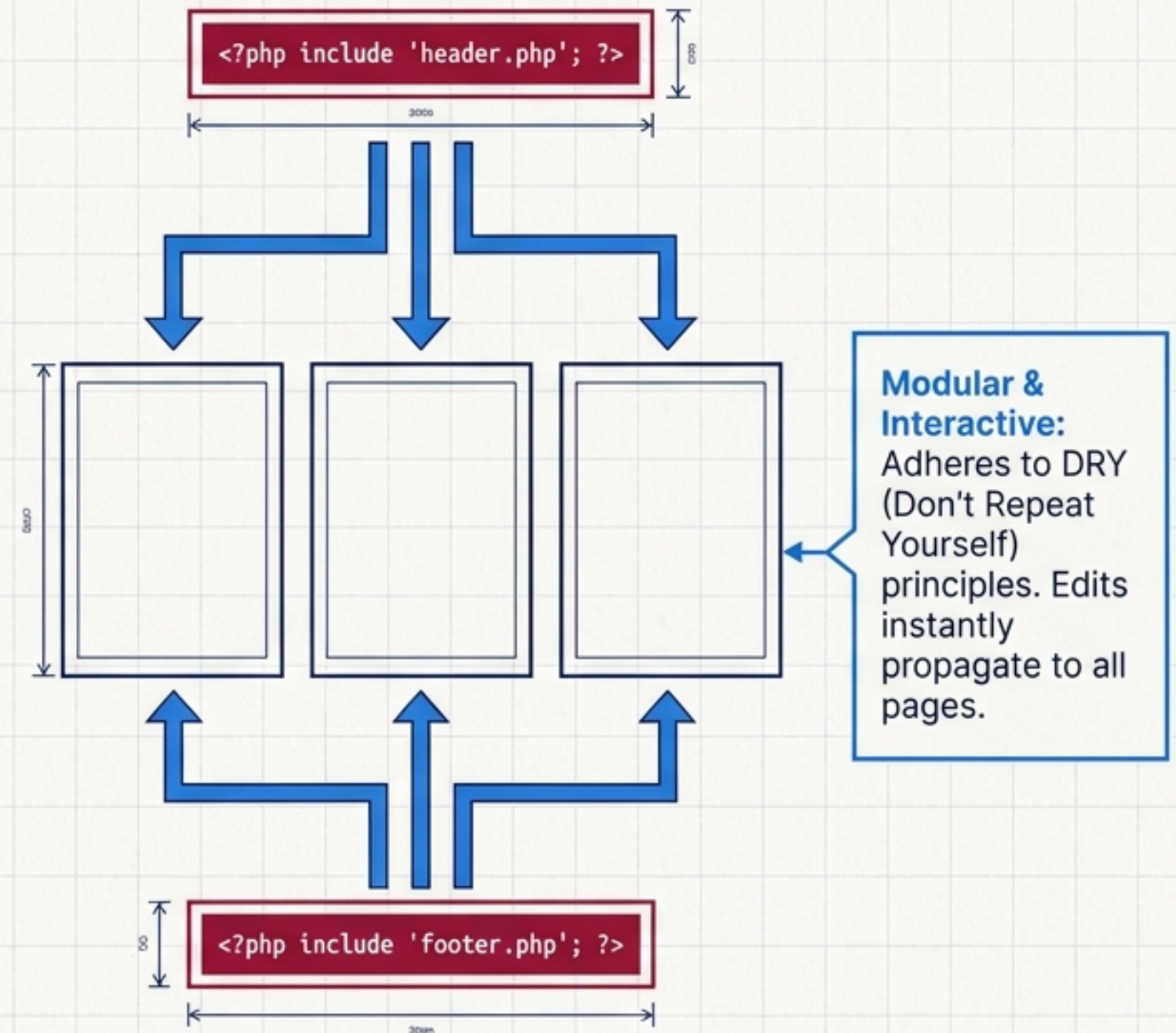
Engineering Dynamic Web Systems

A structural blueprint for integrating PHP and HTML via modular templates, data pipelines, and server-side logic.

Static HTML Architecture



Dynamic PHP Architecture



XAMPP Server / localhost

htdocs/

head.php

footer.php

Reusable Templates

UserForm.php

calculationForm.php

Client Input

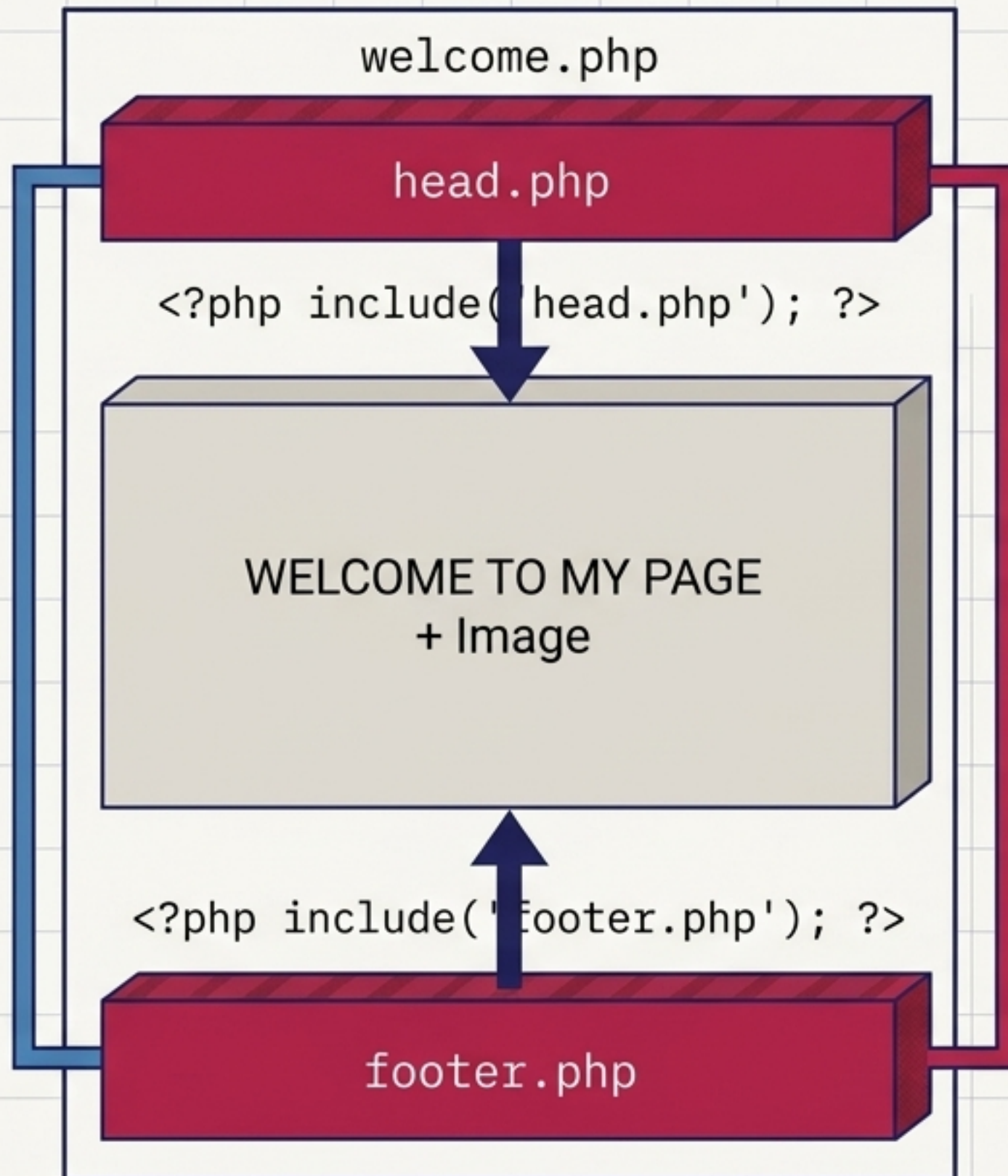
p_form.php

order.php

Processing Engines

A dynamic web application is not a single document; it is a networked directory of specialized files working together on a local server.

The include() Puzzle Box



DITM2123: WEB PROGRAMMING
Name: Your Name
Matric Number: Your Matric Number
[HOME](#) | [FORM](#) | [ORDER FORM](#) | [READ FILE](#) | [WRITE FILE](#) | [VIEW MEMBER](#) | [VIEW PROFILE](#) | [UPDATE PROFILE](#) | [LOGOUT](#)

WELCOME Norazlin Mohammed

First Name	Norazlin
Last Name	Mohammed
Address	Melaka
Email	norazlin@utem.edu.my
Phone Number	0186625588
Gender	Female
Favourite Site	Google Facebook Utem
Comment	good

© norazlin@utem.edu.my 2020 - 2026

One edit to the template files instantly updates every page across the entire application.

Injecting Dynamic Data into Templates

head.php Anatomy

```
<div class="header-container"
style="background-color: #4CAF50;
color: white; padding: 20px; text-
align: center;">
    <p>Date: <span class="dynamic-
date"><?php echo Date('d/m/Y');
?></span></p>
    <h2>DITM2123: WEB
PROGRAMMING</h2>
    <p>Name: Your Name</p>
    <p>Matric Number: Your Matric
Number</p>
    <!-- Navigation Links -->
</div>
```

→ Server fetches
the current
date upon
page load.

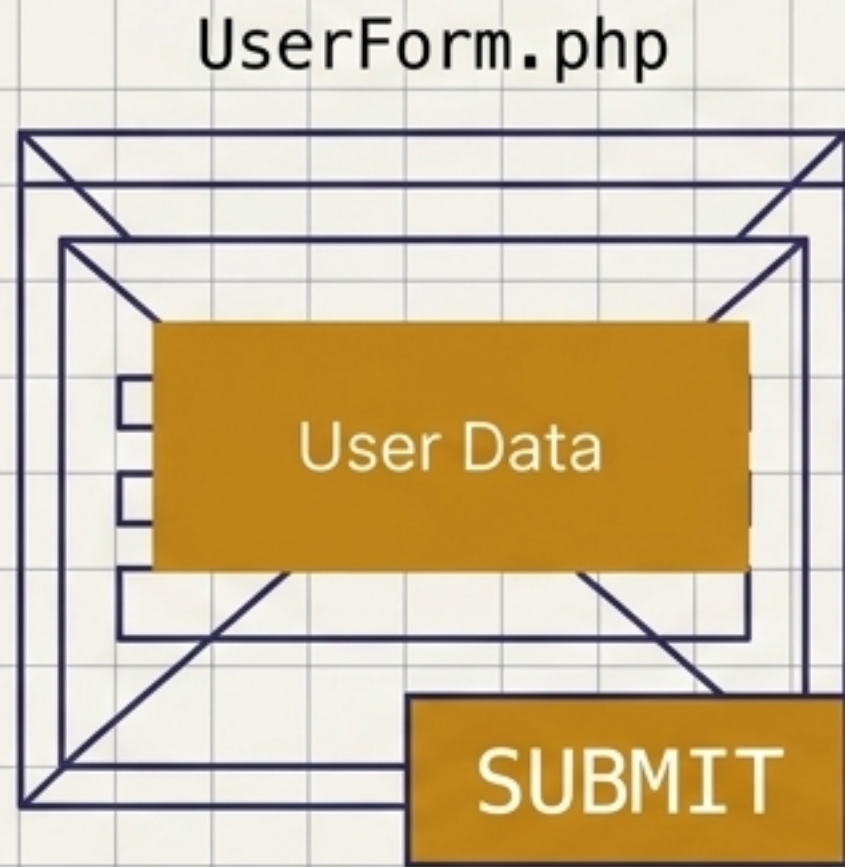
footer.php Anatomy

```
<div class="footer-container"
style="background-color: #4CAF50;
color: white; padding: 10px;
text-align: center; position:
fixed; bottom: 0; width:
100%;">
    <p>&copy;
norazlin@utem.edu.my 2020 -
<span class="dynamic-year">
<?php echo Date('Y'); ?>
</span></p>
</div>
```

→ Dynamically
ensures the
copyright year
is never out of
date.

The Client-Server Boundary

The Client / Browser

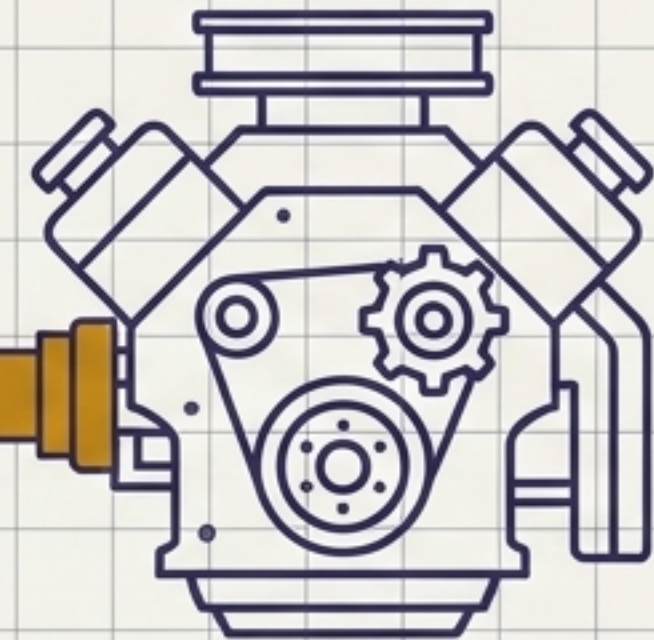


The Pipeline

`$_POST[]` Method

The Server / Localhost

p_form.php



When a user presses submit, the browser packages the input and pushes it across the client-server boundary. PHP catches this package using the `$_POST` superglobal array.

The Input-to-Variable Mapping Matrix

HTML Form Input	Transfer Mechanism	PHP Variable
<code><input name='fname'></code>	<code>\$_POST['fname']</code>	<code>\$fName =</code>
<code><input name='email'></code>	<code>\$_POST['email']</code>	<code>\$email =</code>
<code><input name='gender'></code>	<code>\$_POST['gender']</code>	<code>\$gender =</code>

CRITICAL: The HTML **name attribute** acts as the key. It must perfectly match the string inside the `$_POST['key']` brackets.

Anatomy of a Processing Engine

```
<!DOCTYPE html>
<html>
<head>
  <title>Display User Form Data</title>
  <link rel="stylesheet" type="text/css" href="format.css">
</head>
<body>

  <?php
    include("head.php");

    // Retrieve data from UserForm.php using $_POST
    $fName  = $_POST['fname'];
    $lName  = $_POST['lname'];
    $address = $_POST['address'];
    $email  = $_POST['email'];
    $phone  = $_POST['phone'];
    $gender = $_POST['gender'];
    $site   = $_POST['site'];
    $comment = $_POST['comment'];

    // Display welcome message
    echo "<center><h3>WELCOME " . $fName . " " . $lName . "</h3></center>";
  ?>

  <section>
    <div class="container">
      <div class="login-container">
        <table border="1">
          <tr>
            <th colspan="2">USER INFORMATION</th>
          </tr>
          <tr>
            <td>First Name</td>
            <td><?php echo $fName; ?></td>
          </tr>
          <tr>
            <td>Comment</td>
            <td><?php echo $comment; ?></td>
          </tr>
        </table>
      </div>
    </section>
  </body>
</html>
```

1. Pull in the visual architecture.

This PHP function dynamically inserts the contents of an external file (the header) at this point, establishing the common layout and styles.

2. Catch and store the data payload.

The `$_POST` superglobal captures data submitted from the form. This line specifically extracts the 'address' value and stores it in the `$address` PHP variable.

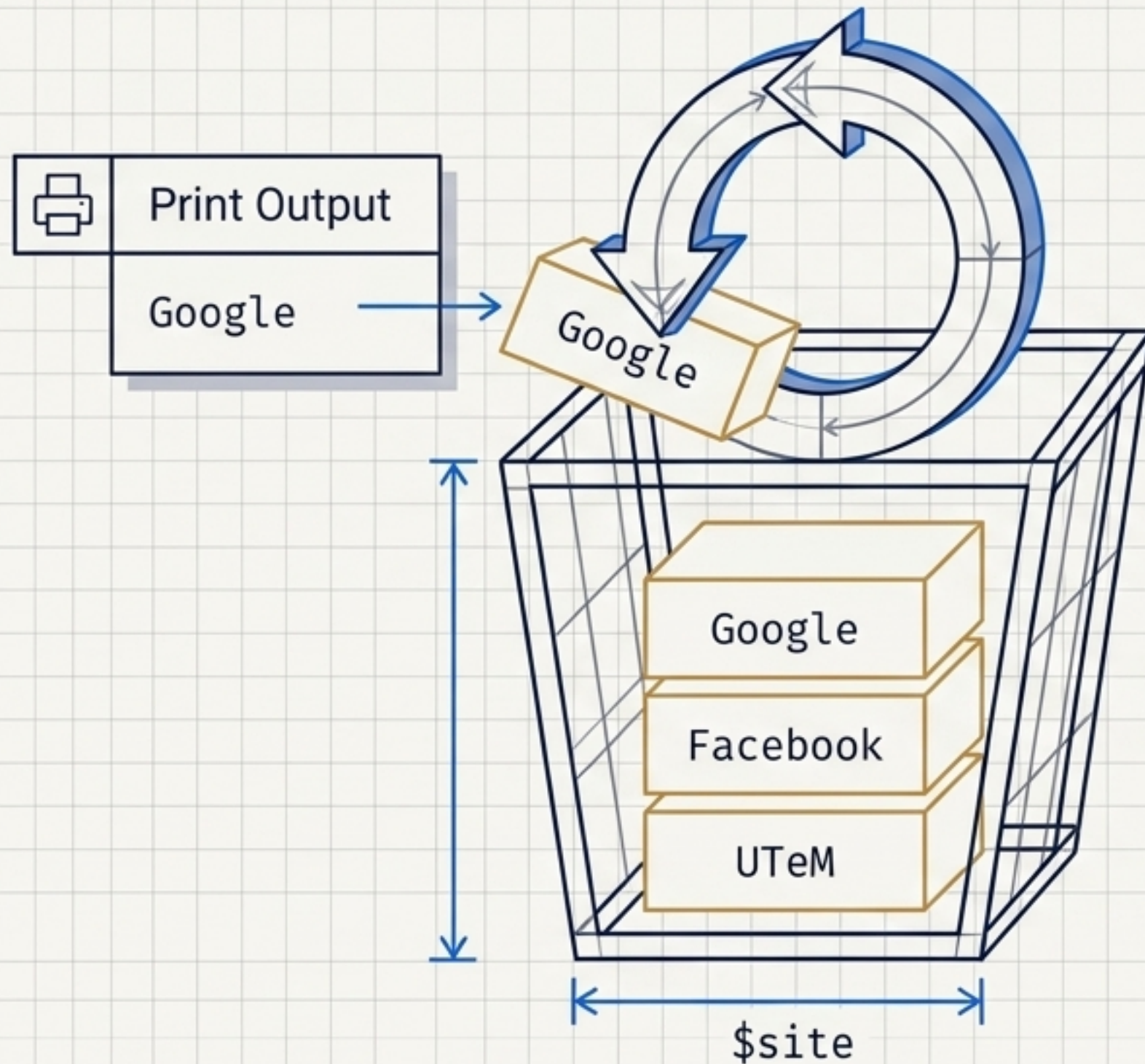
3. Inject the stored variable into an HTML table cell for structured rendering.

PHP's `echo` statement outputs the value held in the `$address` variable directly into the HTML, rendering it within the table cell.

First Name	Mohammed
Last Name	Mohide
Email	itocanlin@olem.edu.my
Phone Number	6 98220080
Gender	Female
Favourite Site	Google
Comment	Feedback

Example Output

Processing Checkboxes: The foreach Loop



Code Implementation

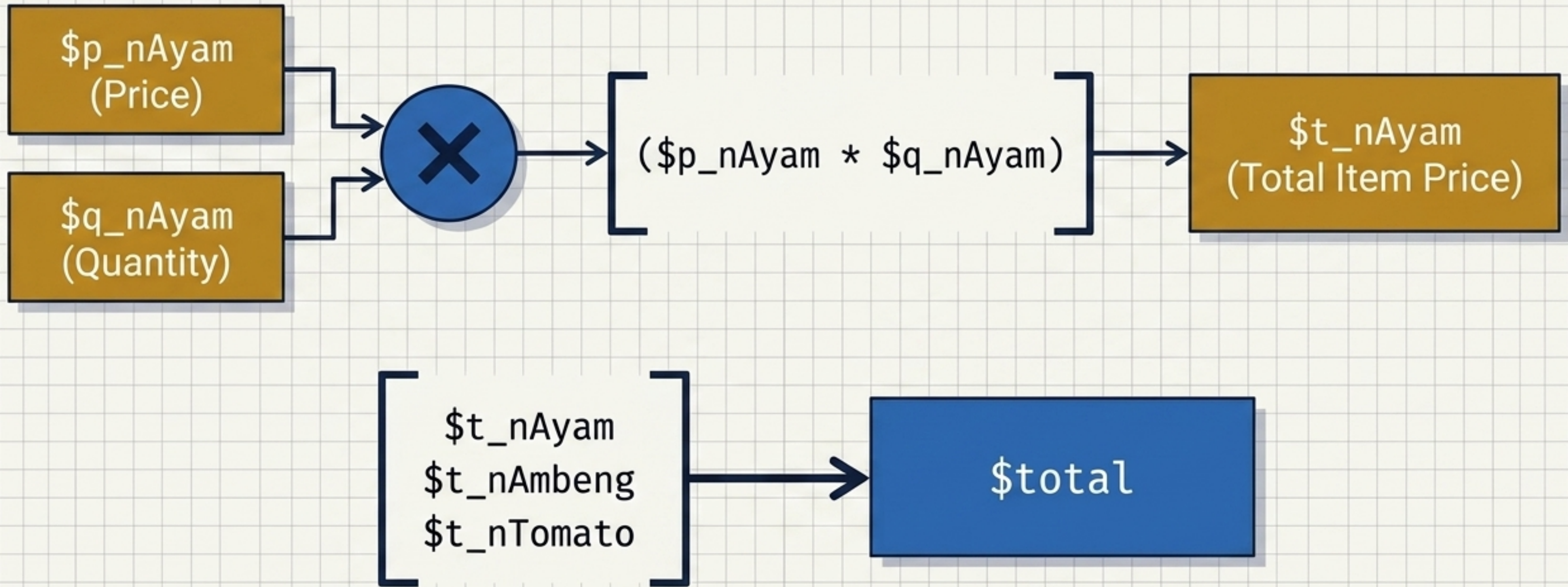
```
<?php
foreach ($site as $value) {
    echo $value . '<br>';
}
```

The basket
(Array of selected
checkboxes)

The single item
currently being
extracted

Prints the item,
adds a line break,
and loops until the
basket is empty.

Server-Side Arithmetic Logic



PHP processes calculations server-side before the browser ever sees the page, allowing secure, real-time total generations for orders.

Refining Raw Mathematical Output

Raw 	Formatted 
\$total = 55.9	\$total = 55.90

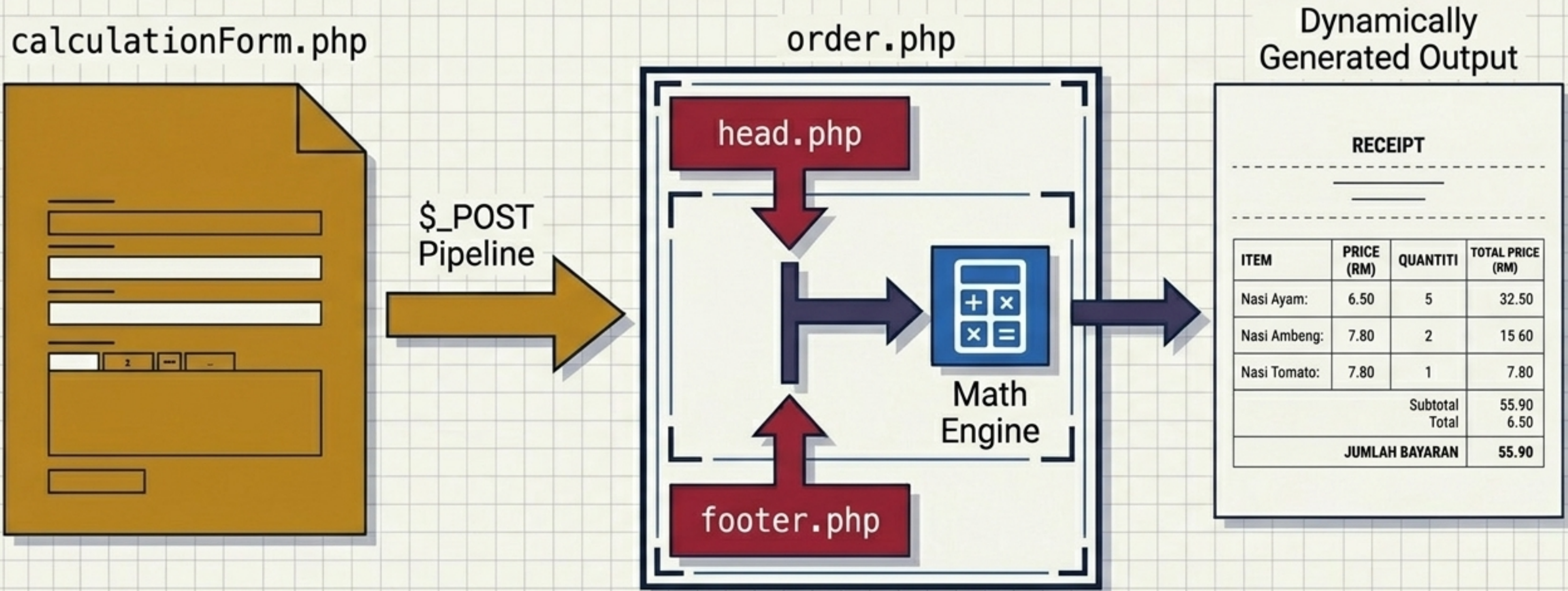
```
// Calculate overall payment
```

```
$total = number_format( ($t_nAyam + $t_nAmbeng + $t_nTomato), 2 );
```

Forces exactly two decimal places for standard currency display.

ITEM:	PRICE (RM)	QUANTITI	TOTAL PRICE (RM)
Nasi Ayam:	6.50	5	32.50
Nasi Ambeng:	7.80	2	15 60
Nasi Tomato:	7.80	1	7.80
JUMLAH BAYARAN	55.90		55.90

The Complete Application Anatomy



You haven't just written scripts; you have engineered an integrated system.
Modularity handles the **UI**, while **POST** handles the data.

Pre-Flight Deployment Checklist

- ☐ Source files placed cleanly inside the htdocs directory folder.
- ☐ System launched via `http://localhost/yourfolder/` (Never open PHP files directly from the desktop).
- ☐ Navigation links between forms and processing pages verified.
- ☐ `include()` files load accurately without path errors.
- ☐ `$_POST` variables display correct submitted data.

Verify outputs, capture execution screenshots, and **compress** the directory for ULearn submission.